Two-frame precleaning

Separating what moves with the star from what stays with the sky, before the velocities are measured

Étienne Artigau

`github.com/eartigau/pca2d_preclean`

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NIRPS & SPIRou teams

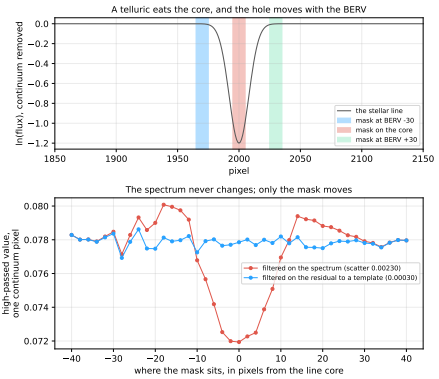
The problem, in one sentence

Two things in every spectrum do not move together

- The **star** is Doppler-shifted by the barycentric motion: tens of km/s over a year.
- The **sky** and the **instrument** are not. Telluric residuals, fringing, persistence: all fixed at the telescope.
- Over a campaign the star therefore *slides past* the contamination.

The lever

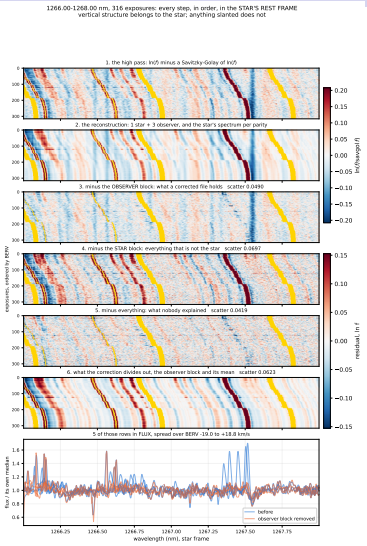
If a feature keeps its place on the detector while the star slides past it, **the feature is not the star**. One block of components anchored in each frame, and the two separate themselves.



A deep line and a mask that walks across it as the BERV changes.

What it looks like in the data

TOI-2120, SPIRou, 316 exposures around 1267 nm, star rest frame



Each row is one exposure, ordered by BERV. Vertical structure belongs to the star; anything slanted

Reading that figure

The same seven panels, in words

1. **The high pass.** Continuum and blaze removed. The *slanted ridges* are the atmosphere walking across the stellar lines as the BERV changes through the year.
2. **The reconstruction**: 1 star + 3 observer components, plus the star's mean spectrum.
3. **Minus the observer block** – this is what a corrected file holds. Scatter $0.0740 \rightarrow 0.0490$, a 34 % drop.
5. **Minus everything**: 0.0419, close to photon noise.

The key visual

In panel 1 the diagonals are obvious. In panel 3 they are gone, and **the vertical lines are untouched**. That is the whole claim of the method, and it is visible without any statistics.

Not a telluric model

P and Q are found by a weighted PCA *on the cube itself*. No line list, no model atmosphere. If the fit says something sits still at the telescope, the campaign showed it sitting still.

The model

Two blocks, one fitted amplitude each, solved per exposure

Let $y(x,t)$ be the high-passed log flux on a grid uniform in $\ln\lambda$:

$$y(x,t) = \underbrace{\bar{S}(x-\delta x_t) + \sum_{i=1}^{n_\star} a_i(t)\, P_i(x-\delta x_t)}_{\text{star frame}} + \underbrace{\sum_{j=1}^{n_\oplus} b_j(t)\, Q_j(x)}_{\text{observer frame}} + \varepsilon$$

- $\bar{S}$: the star's mean spectrum. **Coefficient fixed at exactly one** – the fit can never scale the star away to explain something else.
- P_i: stellar variability. Q_j: the observer block.
- a_i, b_j: the only free parameters.

On a log grid a Doppler shift is a pure translation:

$$\delta x_t = \frac{\operatorname{artanh}(v_t/c)}{\Delta}, \quad v_t = v_{\rm sys} - v_{\rm BERV}(t)$$

Always relativistic. The first-order form is wrong by $\frac{1}{2}(v/c)^2 = 1.5\,\mathrm{m/s}$ at $30\,\mathrm{km/s}$, which is the whole scale of the measurement.

How it is solved

Alternating, each half-step linear and closed-form

1. **Build the cube.** Every exposure resampled onto one common grid with its variance and mask. Nothing co-added.
2. **Take the star's mean** in its rest frame (weighted median down the time axis).
3. **Solve the amplitudes**: with the bases fixed, each exposure is an independent weighted least-squares problem.
4. **Update the bases**: weighted PCA of the residual, each block in its own frame.
5. **Repeat** until χ^2 stops moving.

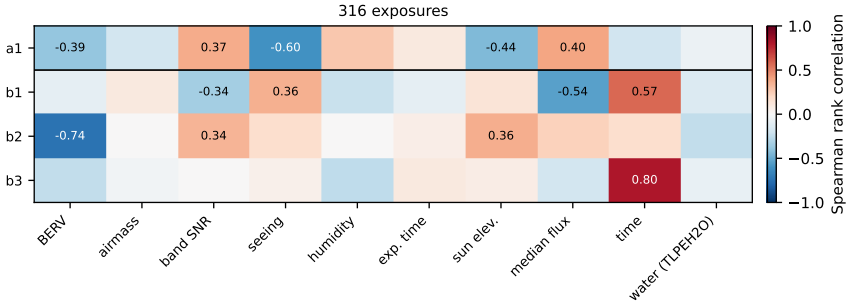
$$\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix} = \left(\mathbf{M}_t^\top \mathbf{W}_t \mathbf{M}_t\right)^{-1} \mathbf{M}_t^\top \mathbf{W}_t \left(\mathbf{y}_t - \bar{S}_t\right)$$

$\mathbf{M}_t$ stacks the shifted P_i and the unshifted Q_j; $\mathbf{W}_t$ is the inverse variance.

Writing $\bar{S}_t$ on the *right-hand side* rather than as a column of $\mathbf{M}_t$ is exactly what pins its coefficient at one.

What the components turn out to be

TOI-2120: rank correlations against everything in the headers



Nothing follows airmass. Largest airmass correlation anywhere: 0.18, and it is on the *stellar* component.

b_3 **follows the clock** at $+0.805$ and nothing else: a monotonic drift over 452 days. Instrument, not weather.

b_2 **follows the BERV** at -0.737, flat in airmass. Not a water column: the residual of an imperfect frame separation.

The high pass: a subtlety that costs m/s

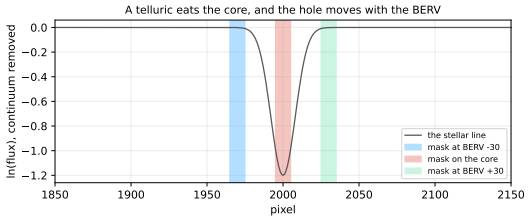
Why the continuum filter is taken against a template

A Savitzky-Golay filter is a local polynomial. It does not know what a line is, so a deep line **drags it down**.

Harmless if it were the same every night. It is not: LBL blanks pixels below e^{-1} transmission, and as the BERV sweeps, **the hole moves across the stellar line**.

The filter is then dragged by a different amount each exposure, leaving a residual that **tracks the BERV without being stellar** – the single most dangerous systematic in this business.

Fix: filter the residual to a median template (which has no holes), then add the template's own filtered version back.



The spectrum is held *exactly* fixed; only the ten-pixel mask moves. Ordinary filter: 0.00230 rms. Against a

A warning about linearity

The identity that looks like it makes this pointless

Savitzky-Golay is linear, so one is tempted to write

$$(s - \bar{S}) - \mathrm{SG}[s - \bar{S}] + \bar{S} - \mathrm{SG}[\bar{S}] \;=\; s - \mathrm{SG}[s]$$

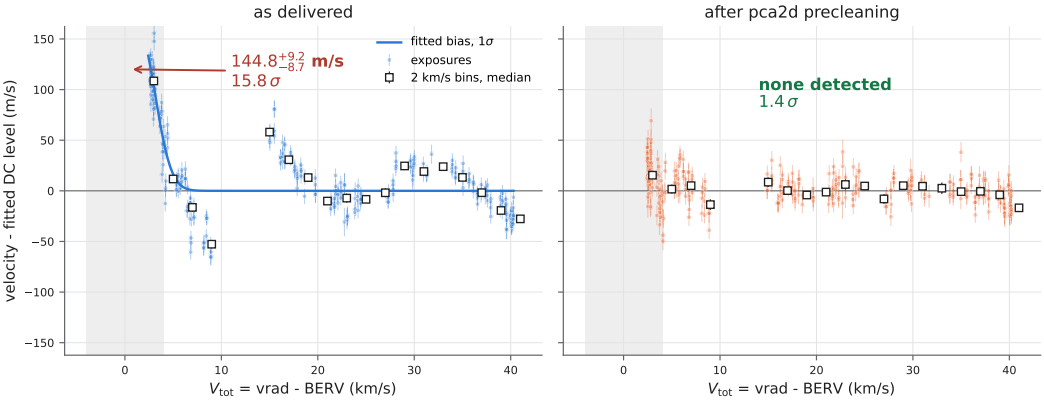
That identity holds only on identical support

The residual $s - \bar{S}$ inherits the exposure's mask, *with holes where the tellurics saturate*. The median template has *no holes*: a pixel blanked tonight was measured on fifty other nights. **Linearity does not survive two different supports**, and the difference between the two sides is exactly the BERV-dependent term.

Measured with a *fixed* mask, the two agree to machine precision – which is how the identity misleads. It took holding the spectrum still and moving *only* the mask to see the effect at all.

The headline result: a bias that follows the BERV

TOI-2120, velocity residual against $V_{\rm tot} = v_{\rm rad} - \mathrm{BERV}$



Near $V_{\rm tot} = 0$ every stellar line sits on the telluric that contaminates it, so anything peaking there is not stellar. Grey band: $|V_{\rm tot}| < 4$ km/s, where the two frames nearly coincide.

The same result, in numbers

A 15.8 σ detection becomes a non-detection

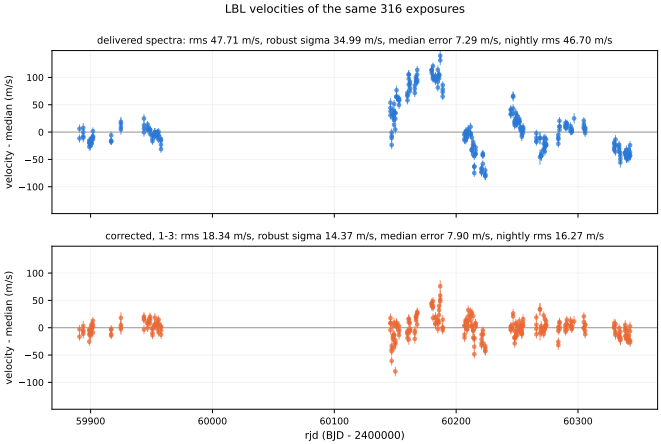
	As delivered	**After precleaning**
Bias peak at $V_{\mathrm{tot}} = 0$	$144.8^{+9.2}_{-8.7}$ m/s	none detected
Width (FWHM)	4.3 km/s	—
Significance	15.8 σ	1.4 σ
$\Delta\mathrm{BIC}$	+416.5	+63.5
95 % upper limit	—	$\lvert\mathrm{peak}\rvert < 216.6$ m/s
Jitter	23.5 m/s	11.7 m/s

Why 4.3 km/s is the right width: that is about a telluric line convolved with the instrumental profile. If the atmosphere makes it, that is the width it should have.

Caveat, stated: amplitude and width are strongly anticorrelated in the posterior ($r = -0.93$), so they are not independently constrained.

Velocities, before and after

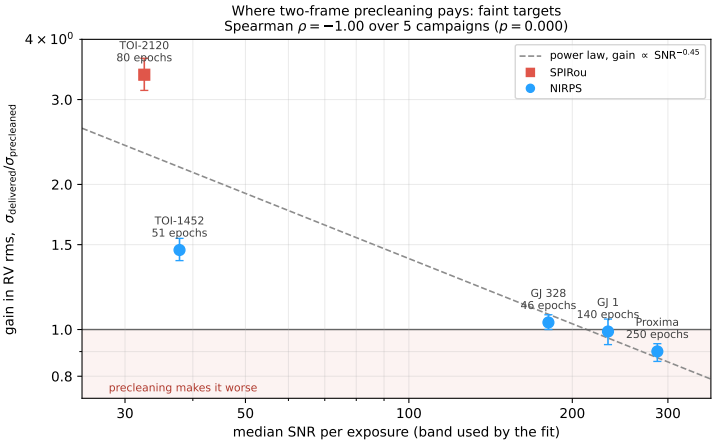
TOI-2120: the same exposures, measured twice by the same LBL stage



Nightly means, 80 shared epochs: 47.98 → 14.20 m/s, a factor 3.38 (paired bootstrap 3.14–3.65). Reduced χ^2: 158.7 → 9.5.

Where it pays, and where it does not

Five campaigns: the gain is ordered by SNR, not by spectral type



Error bars are paired bootstraps over epochs (2000 draws). Perfect rank ordering, but **five points**: a clear trend, not an established law.

The same five campaigns, in numbers

Star	Inst.	SNR	Epochs	rms in	rms out	Gain
TOI-2120	SPIRou	33	80	47.98	14.20	3.38×
TOI-1452	NIRPS	38	51	101.43	69.37	1.46×
GJ 328	NIRPS	181	46	9.34	9.03	1.03×
GJ 1	NIRPS	233	140	2.54	2.56	0.99×
Proxima	NIRPS	287	250	2.96	3.29	0.90×

The honest summary

Faint targets gain; bright ones do not. On a photon-starved spectrum the contamination is comparable to the noise and worth removing. On a bright one it is already below the noise, and fitting it costs more than it returns.

Caveats

Proxima will be **redone** with the HARPS updates; treat that row as provisional. All rows use the nominal 1–3 configuration and one frozen code version.

When not to use it

Failure modes found by measurement, not by argument

Narrow BERV span

The whole separation rests on the star moving relative to the sky.

- Wide span: 2× better
- 0.7 km/s span: 2.4× worse

Two slices of the *same* TOI-4552 campaign, identical otherwise. A fit with no information still returns an answer.

Two stellar components

On TOI-2120 with `-n-star 2` the second amplitude starts following the sky, absorbs a 232 m/s shift, and the rms climbs to 67 m/s.

Shared bases need shared *nights*

Not merely similar SNR. Fitting GJ 1, Proxima and a third target on one basis with only two nights in common made GJ 1 *worse* (3.04 vs 2.61 solo).

What it costs: samples

Every sample the fit could not weight is written as NaN rather than guessed at, and `mask: common` fixes one line set for the whole campaign. On TOI-2120 the files go from **27.2 % NaN as delivered to 51.3 %**. Worth checking on an already photon-limited star.

Practical notes for both teams

Running it `pca2d -object TOI2120 \`

```
-n-star 1 -n-earth 3 \
-stages cube,fit,correct,figures,lbl
```

- Reader picked from `INSTRUME`, never by hand.
- SPIRou, NIRPS, HARPS, HARPS-N, ESPRESSO.
- Corrected spectra on their *original* grids: any velocity code can read them.
- Every run writes its resolved config, the components, a coefficient CSV and a ∼30-page report.

Two traps worth knowing

LBL NIRPS_HE + CADC

A one-line table bug upstream loads the **HA** instrument profile for HE data. Every NIRPS HE run before this was found used the wrong profile.

Telluric fits need a template

Absorption must be fitted on the spectrum *divided by the stellar template*, never raw. The tell-tale: exponents coming out identical with and without a template.

Summary

What it does

- Fits a star-frame block and an observer-frame block *simultaneously*, from the data alone.
- Writes back corrected spectra on their original grids.
- Removes a 144.8 m/s, 15.8 σ BERV-locked bias on TOI-2120, leaving a 1.4 σ non-detection.

Where it pays

- **Faint targets**: 3.4× on TOI-2120, 1.5× on TOI-1452.
- Bright targets: nothing, or slightly worse.

What it needs

- A wide BERV span (tens of km/s).
- Enough epochs to build a clean median template.
- One stellar component, three observer ones, as the nominal starting point.

Open questions

Proxima to be redone with the HARPS updates; the SNR trend rests on five campaigns and wants more; the NaN cost of `mask: common` deserves a closer look on photon-limited stars.

`github.com/eartigau/pca2d_preclean` · `eartigau.github.io/pca2d_preclean`